

Production MLP 拟合结果

$\Delta P^{0.5}$ Stage-3 anchor-off 五 seed 与 seed42 轨迹诊断

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当前模型对象

Production 模型

本 deck 更新为真正的 production MLP : Stage-3 anchor_off · seeds 7/17/42/99/2024 · Stage-2 winner 热启动 · raw-CDF refinement 。

输入与缩放

当前 variant 为 a_dp050_plus_pressures 。输入包含 normalized time 、 tilt 、 plume count 、 injection duration 、 control backpressure ，以及 log injection/chamber pressure 。目标仍使用

$$A = (\Delta P)^{0.5} \rho_a^{-0.25} d_n^{0.5}$$

做物理缩放 。

读法边界

五 seed mean 是论文主结果；seed42 是当前 MLP 五 seed 中 full-clean RMSE 最低的代表性诊断，不替代五 seed 统计 。

Full-Clean 主结果

模型	RMSE	MAE	Bias	P95
Production MLP mean(5)	4.735 $\pm$ 0.037	3.293 $\pm$ 0.039	+0.626	9.167
Production MLP seed42	4.689	3.251	+0.486	9.092
SVGP Stage-3 seed42	4.628	3.173	-0.070	9.242

模型	Cov1	Cov2	mean std
Production MLP mean(5)	0.867	0.985	6.049 mm
Production MLP seed42	0.870	0.986	6.032 mm
SVGP Stage-3 seed42	0.703	0.939	4.000 mm

结论保持克制：MLP 没有在 RMSE/MAE 上超过 SVGP，但它的 coverage 明显更保守。

Source: Thesis/generated/baseline_comparison_20260521/headline_full_clean.csv

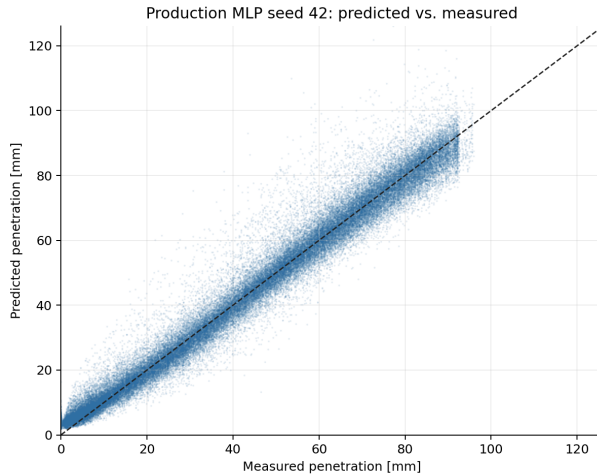
项目	数值
CSV files	2,754
Trajectories	24 316
Prediction points	692 942
Truth range	0.00 mm to 95.88 mm
Time window	0 ms to 5 ms

代表性 seed42

- RMSE = 4.689 mm
- MAE = 3.251 mm
- Bias = +0.486 mm
- P95 absolute error = 9.092 mm
- $1\sigma/2\sigma$ coverage = 0.870 / 0.986

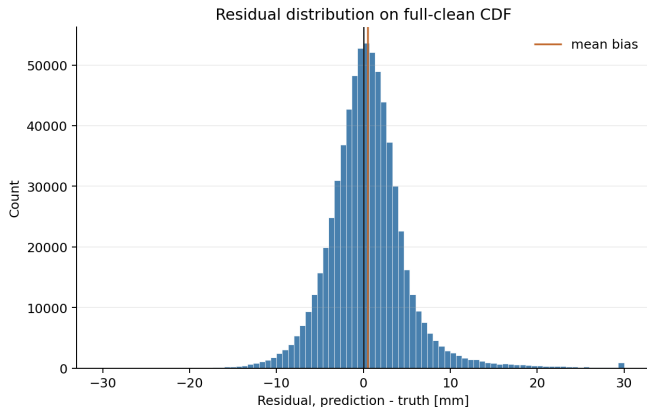
Source: MLP/eval/rmse_eval_clean_20260521_123603_distill_cdf_onset_v2_ablate_anchor_off_20260521_123126/metrics_summary.json

Predicted vs. Measured Penetration



Source: Thesis/images/neural_network_fit_results_20260521/pred_vs_actual_seed42.png

Residual Distribution

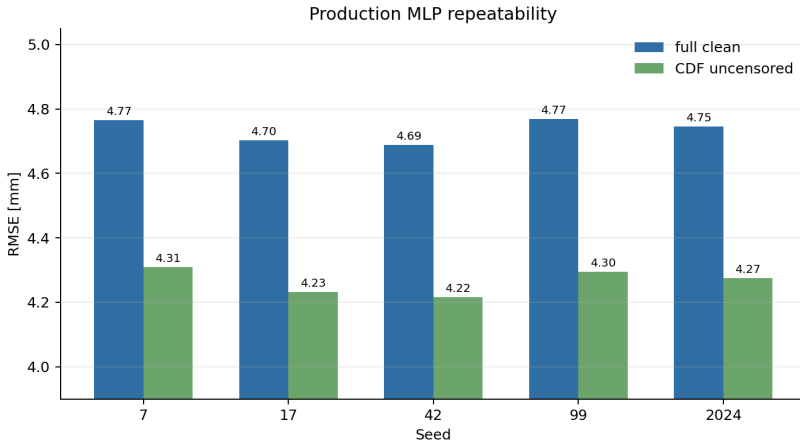


残差定义

$$e = \hat{y}_{\text{pen}} - y_{\text{pen}}.$$

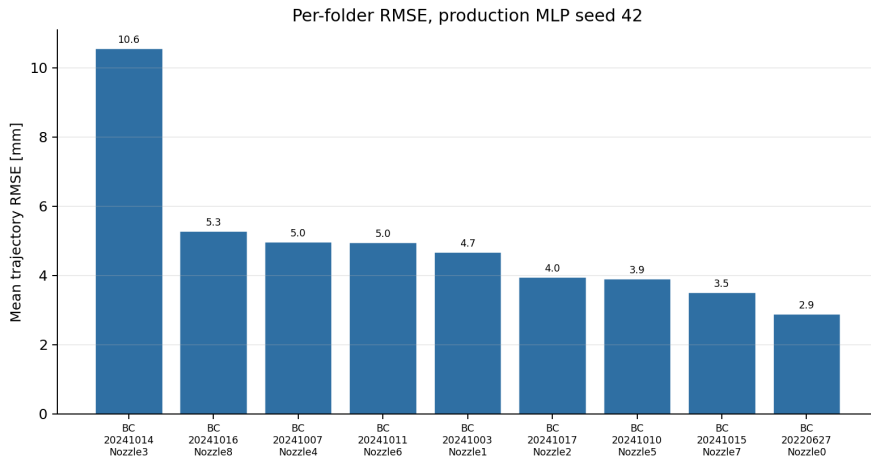
- seed42 bias 为 +0.486 mm，略高估但小于典型 RMSE。
- P95 absolute error 为 9.092 mm。
- 2σ coverage 为 0.986，高于 Gaussian nominal 0.954。

Five-Seed Repeatability



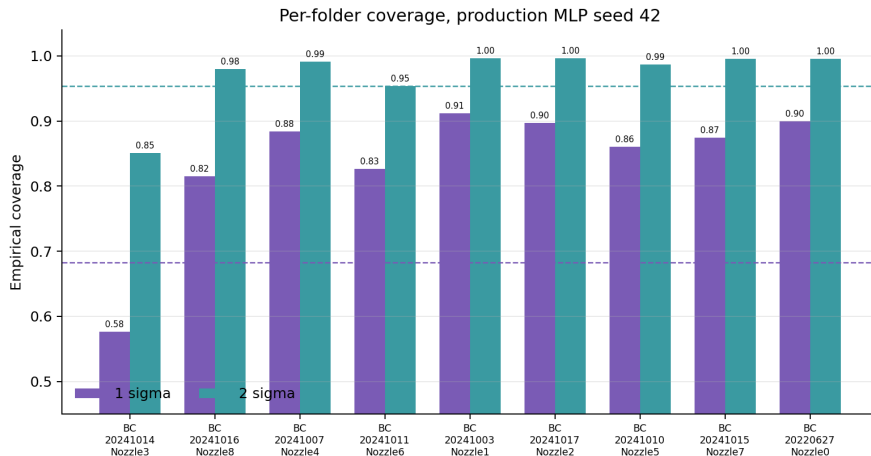
Source: [Thesis/images/neural_network_fit_results_20260521/production_mlp_per_seed_rmse.png](#)

Per-Folder RMSE



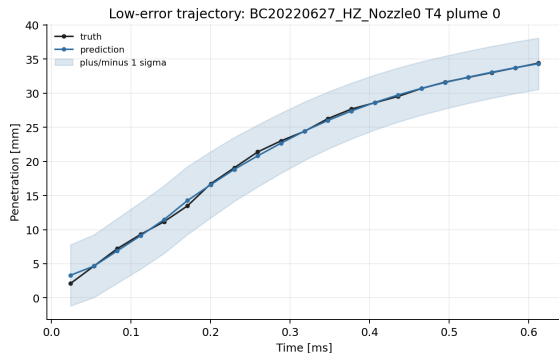
Source: MLP/eval/rmse_eval_clean_20260521_123603_distill_cdf_onset_v2_ablate_anchor_off_20260521_123126/per_folder.csv

Per-Folder Coverage

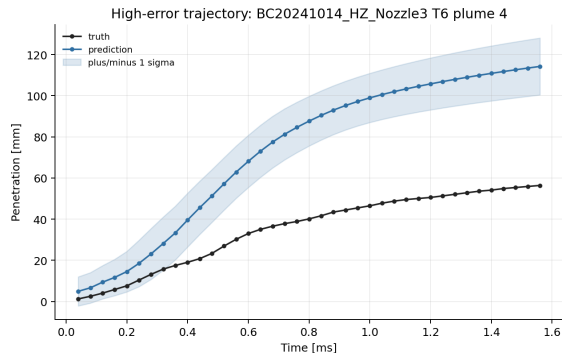


Source: MLP/eval/rmse_eval_clean_20260521_123603_distill_cdf_onset_v2_ablate_anchor_off_20260521_123126/per_folder.csv

代表性轨迹



Nozzle0 T4 plume 0 : RMSE 0.382 mm · coverage 1.00 / 1.00 °



Nozzle3 T6 plume 4 : RMSE 42.763 mm · coverage 0.205 / 0.282 °

Source: Thesis/generated/neural_network_fit_results_20260521/summary.json

- ① 总体拟合已经稳定：五 seed full-clean RMSE 4.735 ± 0.037 mm。
- ② 不确定度偏保守：对 screening surrogate 是可接受方向。
- ③ Nozzle3 仍是主要 failure mode：per-folder RMSE 明显最高，最差轨迹 coverage 也低。
- ④ 不能写成 MLP 已经超过 SVGP：SVGP 的 full-clean RMSE/MAE 仍略好。

论文用语

推荐写法：production MLP closes the gap to SVGP while providing more conservative uncertainty coverage, and it clearly outperforms HA/NS empirical curve baselines.

内容	文件
Slides source	Thesis/slides/neural_network_fit_results_slides_zh/neural_network_fit_results_slides_zh.tex
Figures	Thesis/images/neural_network_fit_results_20260521/
Figure script	Thesis/generated/neural_network_fit_results_20260521/make_figures.py
Representative seed eval	MLP/eval/rmse_eval_clean_20260521_123603_distill_cdf_onset_v2_ablate_anchor_off_20260521_123126/
5-seed aggregate	Thesis/generated/baseline_comparison_20260521/production_mlp_full_clean_per_seed.csv